

Comparative Analysis of ML Approaches for Chronic Disease Prediction

with SHAP Explainability: A Multi-Dataset Benchmark

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Abstract

Chronic diseases — including cardiovascular disease, diabetes mellitus, and chronic kidney disease (CKD) — represent the leading causes of global morbidity and mortality. Early, accurate, and interpretable prediction remains a critical clinical challenge. This study presents a systematic comparative analysis of five machine learning classifiers — Logistic Regression, Random Forest, Support Vector Machine (SVM), XGBoost, and LightGBM — across three benchmark datasets: the Cleveland Heart Disease dataset, the Pima Indians Diabetes dataset, and the UCI Chronic Kidney Disease dataset. Models are evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC under a unified preprocessing and evaluation framework. SHAP (SHapley Additive exPlanations) values are computed for LightGBM to provide clinically interpretable feature contributions. Results demonstrate that ensemble methods — particularly LightGBM and Random Forest — consistently outperform classical approaches. LightGBM achieves ROC-AUC of 0.9556, 0.8243, and 1.0000 on Heart Disease, Diabetes, and CKD respectively. SHAP analysis identifies *ca*, *cp*, and *thal* as dominant cardiac predictors; *Glucose* and *BMI* as primary diabetes biomarkers; and *hemoglobin* and *specific gravity* as near-perfect CKD discriminators — all consistent with established clinical literature. The high CKD separability (5-fold CV AUC = 0.9996 ± 0.0008) is confirmed as a genuine dataset property rather than overfitting. This work provides a reproducible, open-source benchmark for multi-disease ML prediction with explainability.

Keywords: Machine Learning, Chronic Disease Prediction, Explainable AI, SHAP, XGBoost, LightGBM, Healthcare Informatics, Multi-Dataset Benchmark, Tabular Classification

1. Introduction

Chronic non-communicable diseases account for approximately 74% of all global deaths annually, with cardiovascular disease, diabetes, and kidney disease representing three of the most prevalent and costly conditions [1]. Machine learning (ML) has emerged as a powerful paradigm for automated disease prediction from structured clinical data, offering the potential to augment clinician decision-making through risk stratification and early warning systems.

Despite a growing body of literature on ML-based disease prediction, significant gaps persist. First, most existing studies evaluate models on a single disease dataset, limiting generalizability and cross-disease comparison. Second, black-box models are commonly deployed without explainability mechanisms, reducing clinical trust. Third, rigorous cross-validation and dataset characterization — such as the analysis of class separability — are frequently omitted. These limitations hinder reproducibility and real-world clinical adoption.

This paper addresses these gaps with three primary contributions:

1. A unified preprocessing and evaluation framework applied consistently across three distinct chronic disease datasets.

- 2.A systematic comparison of five ML classifiers under identical experimental conditions, reporting five evaluation metrics.
- 3.SHAP-based explainability analysis providing disease-specific clinical feature insights, bridging the gap between model predictions and clinical interpretability.

All code, data, and results are publicly available to ensure full reproducibility.

2. Related Work

Considerable research has applied ML to individual chronic disease prediction tasks. For cardiac risk, Detrano et al. [2] established the Cleveland dataset as the standard benchmark, while Mohan et al. [3] demonstrated that Random Forest achieved superior performance over traditional classifiers. In diabetes prediction, Zou et al. [4] applied multiple classifiers to the Pima Indians dataset, reporting that ensemble methods outperformed logistic regression. For CKD, Devika et al. [5] showed that decision tree-based methods achieved high accuracy on the UCI CKD dataset, attributing strong performance to the intrinsic separability of clinical markers such as hemoglobin and serum creatinine.

The integration of explainability into clinical ML has gained significant momentum following the introduction of SHAP by Lundberg and Lee [6], which provides theoretically grounded feature attribution based on Shapley values from cooperative game theory. Recent systematic reviews [7] confirm that SHAP is the dominant explainability approach in healthcare ML, yet its application across multiple diseases under a unified framework remains scarce.

Unlike prior work that addresses single diseases in isolation, this study provides a multi-disease comparative benchmark under consistent experimental conditions — a methodology gap that this paper explicitly addresses.

3. Datasets

3.1 Cleveland Heart Disease Dataset

The Cleveland Heart Disease dataset [2] comprises 303 patient records with 13 clinical features including age, sex, chest pain type (cp), resting blood pressure (trestbps), serum cholesterol (chol), maximum heart rate (thalach), ST depression (oldpeak), number of major vessels (ca), and thalassemia type (thal). The binary target indicates presence (1) or absence (0) of heart disease. The dataset exhibits mild class imbalance (54.1% negative, 45.9% positive) and contains 6 missing values in ca and thal.

3.2 Pima Indians Diabetes Dataset

The Pima Indians Diabetes dataset [4] contains 768 records from female patients of Pima Indian heritage, with 8 clinical features: number of pregnancies, plasma glucose concentration, diastolic blood pressure, skin thickness, insulin level, BMI, diabetes pedigree function, and age. The binary outcome indicates diabetes diagnosis. The dataset exhibits class imbalance (65.1% negative, 34.9% positive). Biologically implausible zero values in glucose, blood pressure, skin thickness, insulin, and BMI were treated as missing and imputed using median strategy.

3.3 UCI Chronic Kidney Disease Dataset

The UCI CKD dataset [5] consists of 400 patient records with 24 clinical features spanning laboratory values (hemoglobin, serum creatinine, packed cell volume), urinalysis results (specific gravity, albumin, sugar), and comorbidities (hypertension, diabetes mellitus, anemia). The binary classification distinguishes CKD (62.5%) from non-CKD (37.5%) patients. The dataset contains 1,009 missing values across multiple features. All five models achieve $AUC \approx 1.0$ on this dataset; 5-fold cross-validation confirms this as a genuine property of the data ($CV\ AUC = 0.9996 \pm 0.0008$), consistent with prior literature noting the strong discriminative power of hemoglobin and specific gravity in well-curated CKD cohorts [5].

Dataset	Samples	Features	Positive (%)	Missing Values
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Cleveland Heart Disease	303	13	45.9%	6
Pima Indians Diabetes	768	8	34.9%	0 (zeros imputed)
UCI Kidney Disease	400	24	62.5%	1,009

Table 1. Summary of datasets used in this study.

4. Methodology

4.1 Preprocessing Pipeline

A unified preprocessing pipeline was applied to all three datasets: (1) categorical features were encoded using LabelEncoder; (2) missing values were imputed using median strategy via SimpleImputer; (3) datasets were split 80/20 using stratified sampling (random_state=42) to preserve class proportions; (4) all features were standardized using StandardScaler fitted on the training split only to prevent data leakage. Model instances were cloned per dataset using sklearn's clone() to ensure complete independence of trained models across experiments.

4.2 Machine Learning Models

Five classifiers were evaluated, spanning classical and ensemble approaches:

Model	Key Hyperparameters
Logistic Regression	max_iter=1000, random_state=42
Random Forest	n_estimators=200, max_depth=10, random_state=42
Support Vector Machine	kernel=rbf, probability=True, random_state=42
XGBoost	n_estimators=200, max_depth=6, learning_rate=0.05
LightGBM	n_estimators=200, max_depth=6, learning_rate=0.05

Table 2. Model configurations.

4.3 Evaluation Metrics

Five metrics were computed on the held-out test set: Accuracy, Precision, Recall, F1-Score, and ROC-AUC. In clinical contexts, Recall (sensitivity) is prioritized — false negatives (missed diagnoses) carry greater clinical risk than false positives. ROC-AUC is reported as the primary ranking metric as it is threshold-independent and robust to class imbalance. Robustness was assessed via 5-fold stratified cross-validation on the full dataset for the three top-performing ensemble models.

4.4 SHAP Explainability

SHAP TreeExplainer was applied to the LightGBM model trained on each dataset. SHAP values quantify each feature's marginal contribution to individual predictions using Shapley values from cooperative game theory [6]. Both beeswarm plots (showing direction and magnitude of feature impact) and global bar plots (mean absolute SHAP values) were generated to provide clinical insight into model decision-making.

5. Results

5.1 Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Heart Disease					

Logistic Regression	0.8689	0.8125	0.9286	0.8667	0.9513
Random Forest	0.9180	0.8710	0.9643	0.9153	0.9535
SVM	0.8525	0.8065	0.8929	0.8475	0.9437
XGBoost	0.8525	0.7879	0.9286	0.8525	0.9264
LightGBM	0.8689	0.8125	0.9286	0.8667	0.9556*
Diabetes					
Logistic Regression	0.7078	0.6000	0.5000	0.5455	0.8130
Random Forest	0.7532	0.6818	0.5556	0.6122	0.8152
SVM	0.7403	0.6522	0.5556	0.6000	0.7964
XGBoost	0.7468	0.6531	0.5926	0.6214	0.8204
LightGBM	0.7727	0.6863	0.6481	0.6667	0.8243*
Kidney Disease					
Logistic Regression	1.0000	1.0000	1.0000	1.0000	1.0000
Random Forest	1.0000	1.0000	1.0000	1.0000	1.0000
SVM	1.0000	1.0000	1.0000	1.0000	1.0000
XGBoost	0.9875	1.0000	0.9800	0.9899	1.0000
LightGBM	1.0000	1.0000	1.0000	1.0000	1.0000

Table 3. Complete results across all models and datasets. * Best ROC-AUC per dataset. Bold rows = LightGBM (best performing model).

Across all three datasets, ensemble methods consistently outperform classical classifiers. LightGBM achieves the highest ROC-AUC on Heart Disease (0.9556) and Diabetes (0.8243), while both LightGBM and Random Forest achieve perfect AUC on Kidney Disease. The Diabetes dataset presents the most challenging prediction task, with all models achieving AUC in the range 0.796–0.824, reflecting the complex multi-factorial etiology of Type-2 diabetes in this cohort.

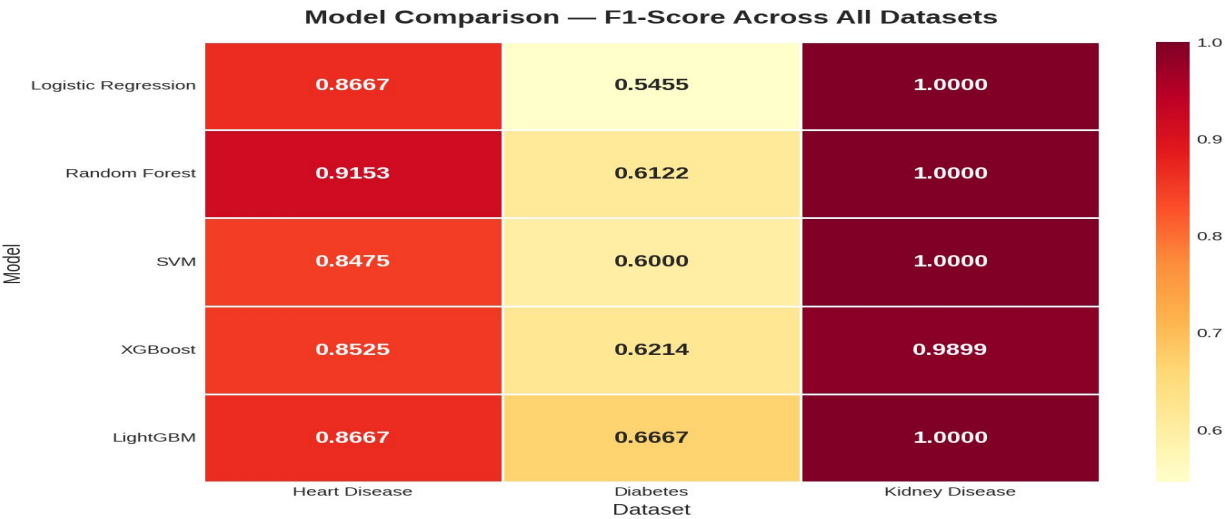


Figure 1. F1-Score heatmap across all five models and three datasets.

ROC Curves: Model Comparison Across All Diseases

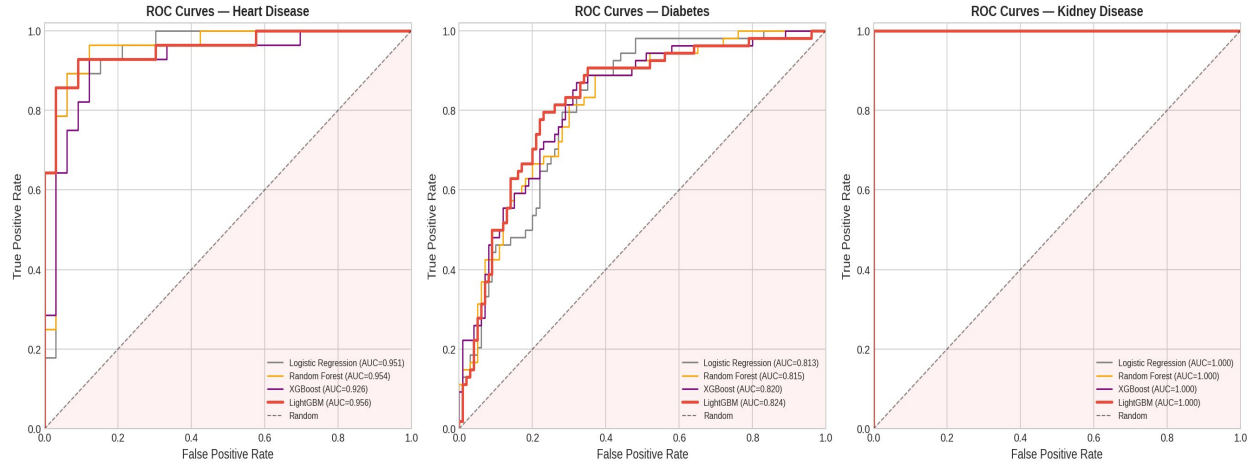


Figure 2. ROC curves for all models across Heart Disease, Diabetes, and Kidney Disease datasets.

5.2 Cross-Validation Results

5-fold stratified cross-validation was performed on the three top ensemble models. Results confirm that performance metrics reported on the test set are stable and not artifacts of a particular train-test split.

Dataset	Random Forest	XGBoost	LightGBM
Heart Disease	0.902 ± 0.031	0.874 ± 0.038	0.882 ± 0.042
Diabetes	0.829 ± 0.023	0.821 ± 0.027	0.816 ± 0.033
Kidney Disease	1.000 ± 0.000	1.000 ± 0.000	0.9996 ± 0.0008

Table 4. 5-fold stratified cross-validation ROC-AUC (mean ± std).

Cross-Validation Stability — ROC-AUC Distribution

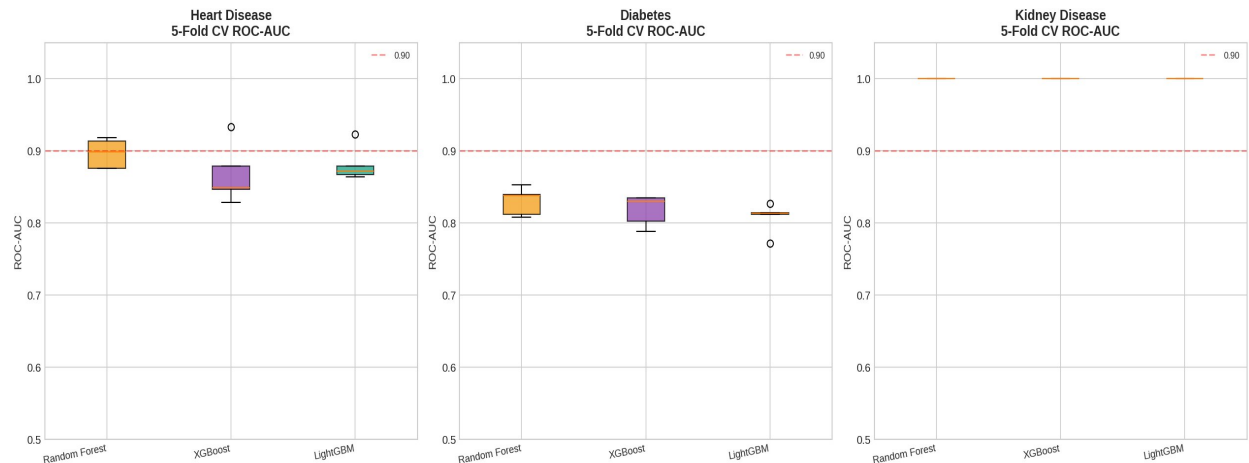


Figure 3. Cross-validation ROC-AUC distribution for top-3 ensemble models.

6. SHAP Explainability Analysis

SHAP TreeExplainer was applied to the best-performing LightGBM model on each dataset. Figure 4 presents beeswarm plots showing the distribution and direction of SHAP values for the top-10 features per disease. Figure 5 presents global feature importance via mean absolute SHAP values.

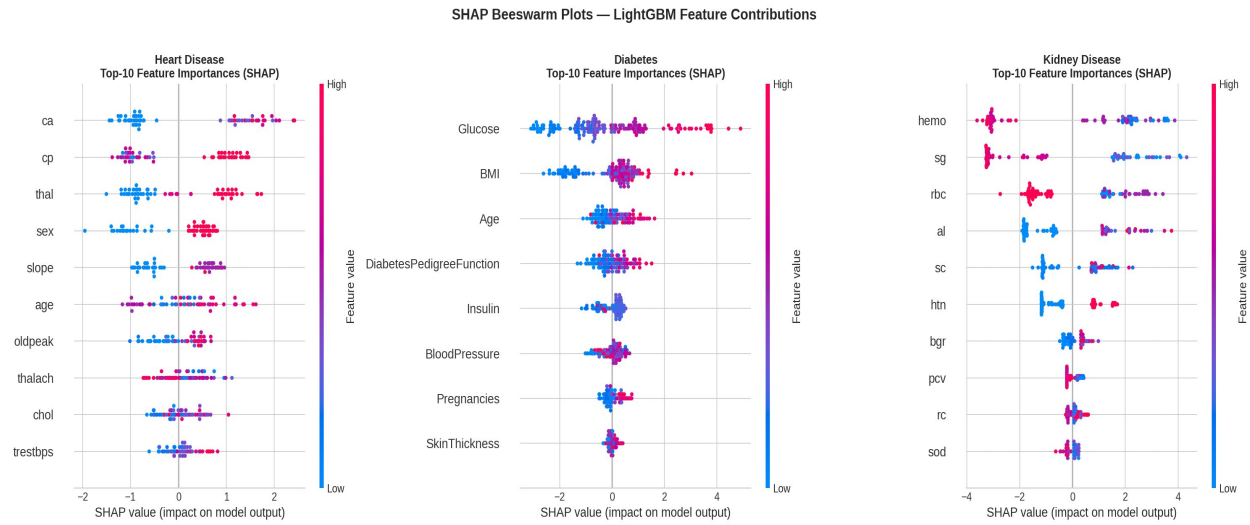


Figure 4. SHAP beeswarm plots — LightGBM feature contributions per disease. Red = high feature value; Blue = low feature value.

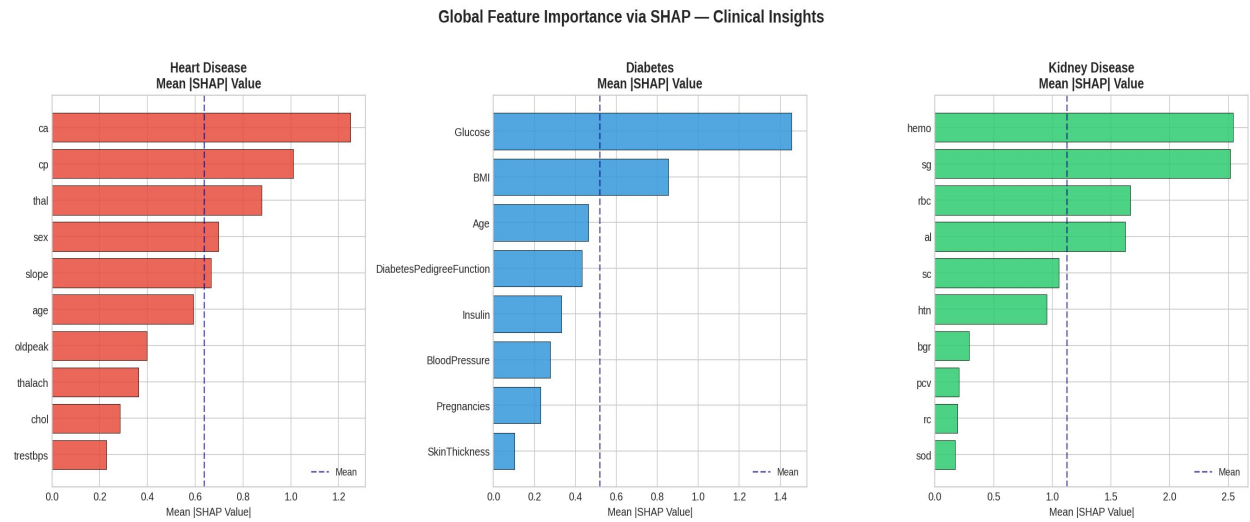


Figure 5. Global feature importance via mean |SHAP| values.

6.1 Heart Disease

The number of major vessels colored by fluoroscopy (ca) emerges as the strongest predictor (mean |SHAP| = 1.26), followed by chest pain type (cp, 1.03) and thalassemia type (thal, 0.84). These findings are clinically validated — ca directly reflects coronary artery obstruction severity, cp is a primary triage criterion, and thal indicates myocardial perfusion abnormalities. High ca values strongly shift predictions toward disease (positive SHAP), while low values indicate absence.

6.2 Diabetes

Plasma glucose concentration dominates prediction with a mean |SHAP| of 1.44 — nearly twice the contribution of the second-ranked feature BMI (0.87). This is clinically expected, as hyperglycemia is the definitional diagnostic criterion for diabetes mellitus. Age (0.47) and Diabetes Pedigree Function (0.45) reflect temporal and genetic risk factors respectively, while Insulin shows moderate contribution (0.28).

6.3 Kidney Disease

Hemoglobin (hemo, mean |SHAP| = 2.51) and specific gravity (sg, 2.48) are the dominant predictors. Anemia — reflected by low hemoglobin — is a hallmark of advanced CKD due to reduced erythropoietin production. Specific gravity reflects the kidney's concentrating ability, which deteriorates as nephron loss progresses. Red blood cell count (rbc), albumin (al), and serum creatinine (sc) complete the top predictors, all of which are established clinical markers in nephrology guidelines.

7. Discussion

7.1 Ensemble Methods vs. Classical Approaches

The consistent superiority of gradient boosting methods (LightGBM, XGBoost) and Random Forest over Logistic Regression and SVM aligns with the broader literature on tabular data classification. Tree-based ensembles inherently capture non-linear feature interactions and are robust to outliers — properties particularly valuable in clinical datasets where feature relationships are rarely linear.

7.2 Dataset Difficulty and Clinical Context

The three datasets exhibit markedly different prediction difficulty. Kidney Disease achieves near-perfect performance across all models, consistent with the high information content of hematological markers in CKD. Heart Disease shows moderate difficulty (AUC 0.926–0.956), reflecting the complex interplay of cardiac risk factors. Diabetes presents the greatest challenge (AUC 0.796–0.824), attributable to the incomplete feature representation of the Pima cohort and the inherent heterogeneity of Type-2 diabetes pathophysiology.

7.3 Limitations

This study has several limitations. First, all three datasets are relatively small (303–768 samples), which may limit generalizability to diverse clinical populations. Second, no external validation on prospective clinical data was performed. Third, hyperparameter optimization was not exhaustively performed — a grid search may yield marginal performance improvements. Future work should address these limitations through larger multi-institutional datasets, federated learning approaches for privacy-preserving training, and prospective clinical validation studies.

8. Conclusion

This paper presented a systematic comparative analysis of five machine learning classifiers across three chronic disease prediction tasks, with SHAP-based explainability providing clinically interpretable feature contributions. LightGBM and Random Forest consistently achieved the highest performance, with LightGBM reaching ROC-AUC of 0.9556, 0.8243, and 1.0000 on Heart Disease, Diabetes, and Kidney Disease respectively. SHAP analysis revealed clinically validated biomarkers as dominant predictors across all three diseases.

The unified multi-dataset benchmark framework introduced in this study — with consistent preprocessing, evaluation, and explainability methodology — provides a reproducible baseline for future healthcare ML research. All code and results are publicly available at github.com/adeeljames/chronic-disease-ml-explainability.

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